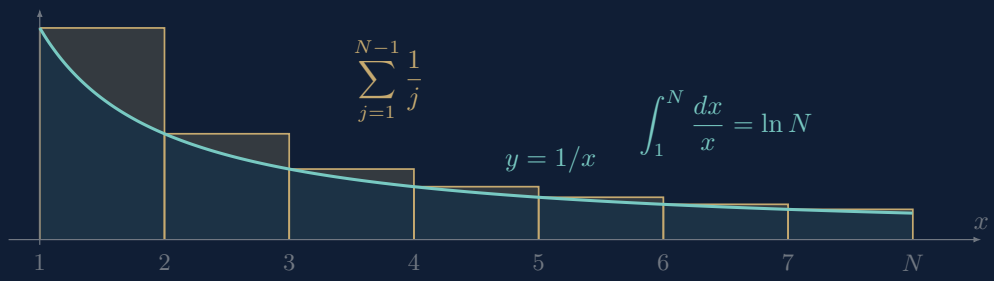
Mathematics Autumn 2026

# Calculus III

MATH 1953 · CRN 1114 · Undergraduate

Calculus is the study of functions on the continuum, their variations, their summations, and other such notions. The fundamental tool of calculus is the idea of limits and the rigorous notion of approximation. It is the foundation of all modern science, the language which we use to model the world, and the greatest discovery by the greatest minds. This course introduces sequences and their summations, called series. We also study Taylor polynomials as approximations.




Euler's constant

$$\lim_{N\to\infty} \left( \sum_{j=1}^{N} \frac{1}{j} - \ln N \right) = \gamma \approx 0.5772156649\ldots$$

## Topics

- Sequences and their limits
- Monotone convergence theorem
- Bernoulli–L'Hôpital theorem and Taylor theorems
- Numerical series
- Comparison with integrals, direct comparison, limit comparison
- Absolute convergence, alternating series test, ratio and root tests
- Power series and Taylor series

## Prerequisites

Calculus I and II

## Instructor

Prof. Frédéric Latrémolière

## Meetings

Tuesdays & Thursdays
10:00–11:50 a.m.
CC 150

September 8–November 16

## TA session

Fridays
11:00–11:50 a.m.
Olin 205